

EDA-Data Cleaning

Code:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

df = pd.read_csv('test_Y3wMUE5_7gLdaTN.csv')

print("Initial Data Overview:")
print(df.info())

print("\nMissing Values in Each Column:\n", df.isnull().sum())
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
plt.title("Missing Value Heatmap")
# plt.show() # Disabled to allow non-interactive execution
print("Heatmap generated, but plt.show() skipped for non-interactive execution.")

for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
    df[col] = df[col].fillna(df[col].mode()[0])

df['LoanAmount'] = df['LoanAmount'].fillna(df['LoanAmount'].median())
df['Loan_Amount_Term'] = df['Loan_Amount_Term'].fillna(df['Loan_Amount_Term'].mode()[0])
df['Credit_History'] = df['Credit_History'].fillna(df['Credit_History'].mode()[0])

initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
print(f"\nRemoved {initial_rows - df.shape[0]} duplicate rows.")

df['Dependents'] = df['Dependents'].replace('3+', 3).fillna(0).astype(int)

for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
    df[col] = df[col].astype(str).str.strip().str.capitalize()

min_max_scaler = MinMaxScaler()
scale_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])

scaler = StandardScaler()
df[['Credit_History']] = scaler.fit_transform(df[['Credit_History']])

print("\nCleaned Data Summary:")
print(df.info())
print(df.head())
```

Output:

```
Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 370 entries, 0 to 369
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                370 non-null   object
1   Gender                 352 non-null   object
2   Married                346 non-null   object
3   Dependents             359 non-null   object
4   Education              370 non-null   object
5   Self_Employed          348 non-null   object
6   ApplicantIncome        370 non-null   int64
7   CoapplicantIncome      370 non-null   int64
8   LoanAmount             369 non-null   float64
9   Loan_Amount_Term       333 non-null   float64
10  Credit_History          336 non-null   float64
11  Property_Area           370 non-null   object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.8+ KB
None
```

Missing Values in Each Column:

```
Loan_ID      0
Gender       18
Married      24
Dependents   11
Education    0
Self_Employed 22
ApplicantIncome 0
CoapplicantIncome 0
LoanAmount   1
Loan_Amount_Term 37
```

```
Property_Area      0
dtype: int64
Heatmap generated, but plt.show() skipped for non-interactive execution.
```

Removed 3 duplicate rows.

Cleaned Data Summary:

```
<class 'pandas.core.frame.DataFrame'>
Index: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                367 non-null   object
1   Gender                 367 non-null   object
2   Married                367 non-null   object
3   Dependents             367 non-null   int64
4   Education              367 non-null   object
5   Self_Employed          367 non-null   object
6   ApplicantIncome        367 non-null   float64
7   CoapplicantIncome      367 non-null   float64
8   LoanAmount             367 non-null   float64
9   Loan_Amount_Term       367 non-null   float64
10  Credit_History          367 non-null   float64
11  Property_Area           367 non-null   object
dtypes: float64(5), int64(1), object(6)
memory usage: 37.3+ KB
None
```

	Loan_ID	Gender	Married	...	Loan_Amount_Term	Credit_History	Property_Area
0	LP001000	Male	No	...	360.0	0.527964	Rural
1	LP001001	Male	Yes	...	360.0	0.527964	Urban
2	LP001002	Male	Yes	...	480.0	0.527964	Urban
3	LP001003	Male	Yes	...	12.0	0.527964	Urban
4	LP001004	Male	Yes	...	300.0	-1.894070	Urban

[5 rows x 12 columns]